Learning for Data Science

Course 12: Feature Selection

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Outline

Similarity based methods: Similarity technique

Information Theory based methods

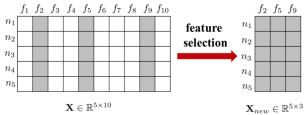
Sparse learning based methods: What is Feature Sparsity?

Statistical methods

Feature Selection with Structured Features: A Framework

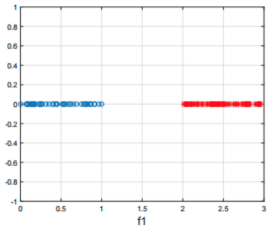
What is feature selection?

- A procedure in machine learning to find a subset of features that produces 'better' model for given dataset
  - Avoid overfitting and achieve better generalization ability
  - Reduce the storage requirement and training time
  - Interpretability

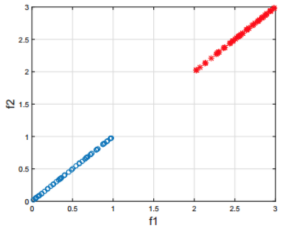


Relevant vs redundant features

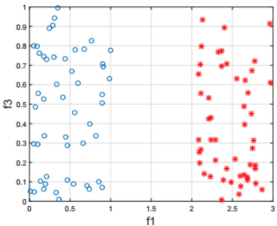
- Feature selection keeps relevant features for learning and removes redundant and irrelevant features
- For example, for a binary classification task (f_1 is **relevant**; f_2 is **redundant** given f_1; f_3 is **irrelevant**)



(a) relevant feature f_1



(b) redundant feature f_2

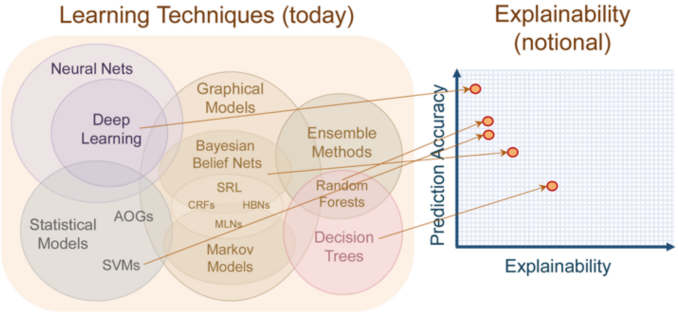


(c) irrelevant feature f_3

Feature Extraction vs. Feature Selection

- **Commonalities**
  - Speed up the learning process
  - Reduce the storage requirements
  - Improve the learning performance
  - Build more generalized models
- **Differences**
  - Feature extraction obtains new features while feature selection chooses a subset of original ones
  - Feature selection maintains physical meanings and gives models better readability and interpretability

Interpretability of Learning Algorithms



http://nautil.us/issue/40/learning/is-artificial-intelligence-permanently-inscrutable

With feature selection, both the accuracy and interpretability of most learning algorithms can be enhanced !

When is feature selection important?

- Noisy data
- Lots of less frequent features
- Use multi-type features (e.g. multi-modal)
- Too many features compared to samples
- Complex model
- Samples in real scenario are different with training & test samples

Feature Selection Algorithms

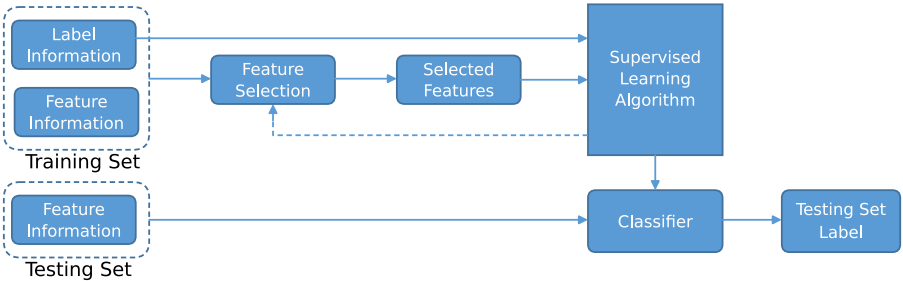
- From the **label perspective** (whether label information is involved during the selection phase):
  - Supervised
  - Unsupervised
  - Semi-Supervised

Feature Selection Algorithms

- From the **label perspective** (whether label information is involved during the selection phase):
  - Supervised
  - Unsupervised
  - Semi-Supervised
- From the **selection strategy perspective** (how the features are selected):
  - Wrapper methods
  - Filter methods
  - Embedded methods

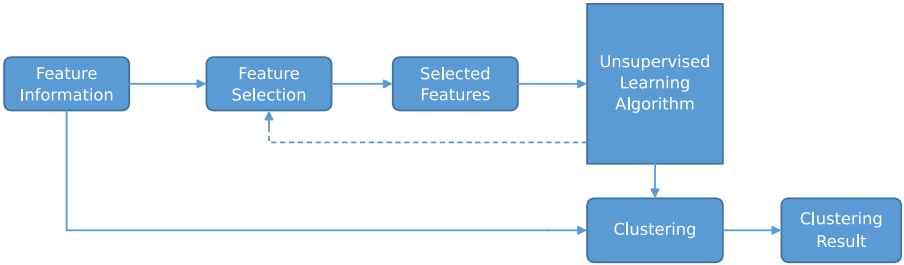
Supervised Feature Selection

- Supervised feature selection is often for classification or regression problems
- Find discriminative features that separate samples from different classes (classification) or approximate target variables (regression)



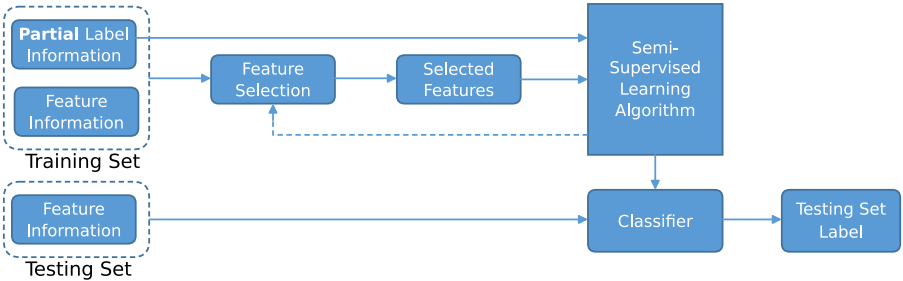
Unsupervised Feature Selection

- It is often for clustering problems
- Label information is expensive to obtain which requires both time and effort
- Unsupervised methods seek alternative criteria to define feature relevance

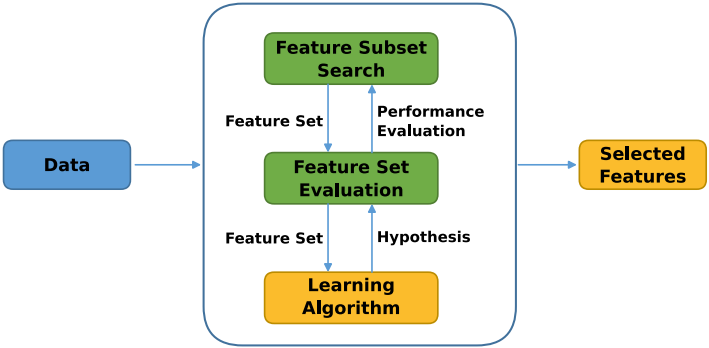


Semi-Supervised Feature Selection

- Often there is little labeled data and a large amount of unlabeled data
- Semi-supervised methods exploit both labeled and unlabeled data to find relevant features



Wrapper Methods



- Step 1: search for a subset of features
- Step 2: evaluate the selected features
- Repeat Step 1 and Step 2 until stopped

Feature Selection Techniques

- **Subset selection method : Forward Search and Backward Search**
  - **Forward Search**
    - Start with no features
    - Greedily include the most relevant feature
    - Stop when selected the desired number of features
  - **Backward Search**
    - Start with all the features
    - Greedily remove the least relevant feature
    - Stop when selected the desired number of features
  - Inclusion/Removal criteria uses **cross-validation**

Wrapper Methods

- Can be applied for **ANY model.**
- Rely on the predictive **performance of a predefined learning algorithm** to assess features
- **Shrink / grow feature** set by greedy search
- Repeat until **some stopping criteria** are satisfied
- Achieve **high accuracy** for a particular learning method
- Run **Cross Validation / train-val split per feature**
- **Computationally expensive** (worst case search space is 2^d), some typical search strategies are
  - Sequential search
  - Best-first search
  - Branch-and-bound search

Filter Methods

- **Independent** of any learning algorithms
- **Rely on certain characteristics** of data to assess feature importance (e.g., feature correlation, mutual information...)
- More **efficient** than wrapper methods
- The selected features **may not be optimal** for a particular learning algorithm or problem.



Feature Selection Techniques

- **Single feature evaluation: measure feature quality by all kinds of metrics**
  - Frequency based
  - Dependence of feature and label (co-occurrence), e.g., *Mutual information, Chi square statistic*
  - Information theory, KL divergence, Information gain
  - Gini indexing

Embedded Methods

- A trade-off between wrapper and filter methods by embedding feature selection into the model learning, e.g., ID3
- Inherit the merits of wrapper and filter methods
  - Include the interactions with the learning algorithm
  - More efficient than wrapper methods
- Like wrapper methods, they are biased to the underlying learning algorithms

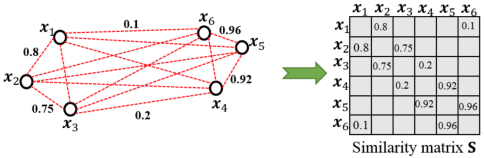


Selection Criteria

- **Traditional feature selection**
  - Similarity based methods
  - Information Theory based methods
  - Sparse learning based methods
  - Statistical methods

Similarity based methods: Similarity technique

- Pairwise data similarity is often encoded in the data similarity matrix



- E.g., without class label information, it can be defined by the RBF kernel

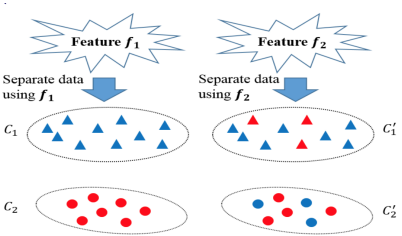
$$\mathbf{S}_{ij} = e^{-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}}$$

- E.g., using the class labels, the similarity can be obtained as

$$\mathbf{S}_{ij} = \begin{cases} \frac{1}{n_l} & \text{if } y_i = y_j = l \\ 0 & \text{otherwise} \end{cases}$$

Similarity based Feature Selection

- Similarity based methods assess the importance of features by their ability to preserve data similarity
- A good feature should not randomly assign values to data instances
- A good feature should assign similar values to instances that are close to each other – (the "closeness" is obtained from data similarity matrix)
- Different shapes denote different values assigned by a feature



Similarity based Methods – A General Framework

- Given the data similarity matrix $\mathbf{S}$, the most relevant features $\mathcal{S}$ maximize:

$$\underbrace{\max_{\mathcal{S}} U(\mathcal{S})}_{\text{utility of feature set } \mathcal{S}} = \max_{\mathcal{S}} \sum_{f \in \mathcal{S}} \underbrace{U(f)}_{\text{utility of feature } f} = \max_{\mathcal{S}} \sum_{f \in \mathcal{S}} \hat{\mathbf{f}}' \hat{\mathbf{S}} \hat{\mathbf{f}}$$

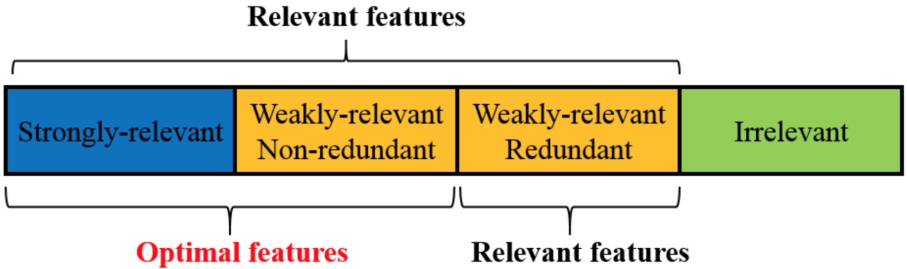
- Utility $U(\cdot)$: how well the feature set preserves the data similarity structure
- It is often solved by greedily selecting the top features that maximize their individual utility $U(f)$
- Different methods vary in the way how the vector $\mathbf{f}$ and similarity matrix $\mathbf{S}$ are transformed to $\hat{\mathbf{f}}$ and $\hat{\mathbf{S}}$

Similarity based Methods Summary

- Many others can also be reduced to the general similarity based feature selection framework
  - Batch-mode Laplacian score (Nie *et al.*, 2008)
  - RelieF (Robnik-Šikonja and Kononenko, 2003)
  - HSIC Criterion (Song *et al.*, 2007) ...
- Pros:
  - Simple and easy to calculate the feature scores
  - Selected features can be generalized to subsequent learning tasks
- Cons:
  - Most methods cannot handle feature redundancy

Information Theory based methods

- Exploit different heuristic filter criteria to measure the importance of features
- The target is to find these "optimal" features



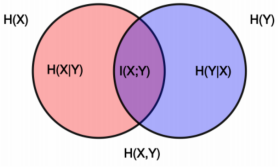
Preliminary - Information Theoretical Measures

- Entropy of a discrete variable X:

$$H(X) = -\sum_{x_i \in X} P(x_i) \log P(x_i)$$

- Conditional entropy of X given Y:

$$H(X|Y) = -\sum_{y_j \in Y} P(y_j) \sum_{x_i \in X} P(x_i|y_j) \log P(x_i|y_j)$$



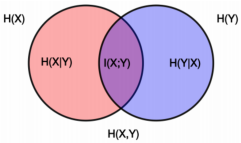
Preliminary - Information Theoretical Measures

- Information gain between X and Y:

$$I(X;Y) = H(X) - H(X|Y) = \sum_{x_i \in X} \sum_{y_j \in Y} P(x_i, y_j) \log \frac{P(x_i, y_j)}{P(x_i)P(y_j)}$$

- Conditional information gain:

$$\begin{aligned} I(X;Y|Z) &= H(X|Z) - H(X|Y,Z) \\ &= \sum_{z_k \in Z} P(z_k) \sum_{x_i \in X} \sum_{y_j \in Y} P(x_i, y_j|z_k) \log \frac{P(x_i, y_j|z_k)}{P(x_i|z_k)P(y_j|z_k)} \end{aligned}$$



Information Theoretic based Methods - A General Framework

- Searching for the best feature subset is NP-hard, most methods employ forward/backward sequential search heuristics
- E.g., for forward search, given selected features $\mathcal{S}$, the next feature f_k should:
  - Maximize its correlation with class labels Y: $I(f_k; Y)$
  - Minimize the redundancy w.r.t. selected features in $\mathcal{S}$: $\sum_{f_j \in \mathcal{S}} I(f_j; f_k)$
  - Maximize its complementary info w.r.t. selected features in $\mathcal{S}$: $\sum_{f_j \in \mathcal{S}} I(f_j; f_k | Y)$

Information Theoretic based Methods - A General Framework

- Given selected features $\mathcal{S}$, the feature score for the next selected feature can be determined by

$$score(f_k) = I(f_k; Y) + \sum_{f_j \in \mathcal{S}} g\left[I(f_j; f_k), I(f_j; f_k|Y)\right]$$

  - $g(*)$: a function
  - The higher the feature score, the better the selected feature is

Information Theoretic based Methods - A General Framework

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  - $g(*)$: a function
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- If $g(*)$ is a linear function, then it can be represented as

$$score(f_k) = I(f_k; Y) - \beta \sum_{f_j \in \mathcal{S}} I(f_j; f_k) + \lambda \sum_{f_j \in \mathcal{S}} I(f_j; f_k|Y)$$

  - β and λ are between 0 and 1

- In general, $g(*)$ can be a nonlinear function

Mutual Information Feature Selection (Battiti, 1994)

- Information gain only considers feature relevance
- Features also should not be redundant to each other
- The score of a new unselected feature f_k:

$$score(f_k) = \underbrace{I(f_k; Y)}_{\text{maximize feature relevance}} - \beta \underbrace{\sum_{f_j \in \mathcal{S}} I(f_k; f_j)}_{\text{minimize feature redundancy}}$$

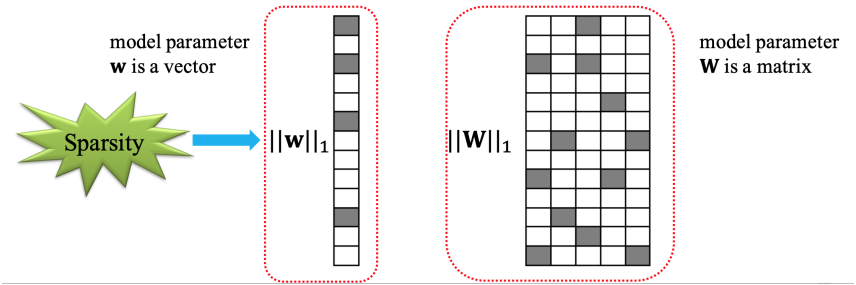
- It is also a special case of the linear function by setting $\lambda = 0$

Information Theoretical based Methods - Summary

- Other information theoretical based methods
  - Fast Correlation Based Filter (Yu and Liu, 2004)
  - Interaction Gain Feature Selection (El Akadi *et al.*, 2008)
  - Conditional MIFS (Cheng *et al.*, 2011)...
- Pros:
  - Can handle both feature relevance and redundancy
  - Selected features can be generalized for subsequent learning tasks
- Cons:
  - Most algorithms can only work in a supervised scenario
  - Can only handle discretized data

Sparse learning based methods: What is Feature Sparsity?

- The model parameters in many data mining tasks can be represented as a vector $\mathbf{w}$ or a matrix $\mathbf{W}$
- Sparsity indicates that many elements in $\mathbf{w}$ and $\mathbf{W}$ are small or exactly zero



Sparse Learning Methods - A General Framework

- Start from the binary classification or the univariate regression problem
- Let $\mathbf{w}$ denote the model parameter (a.k.a. feature coefficient), obtained by solving

$$\mathbf{w} = \arg\min_{\mathbf{w}} \; loss(\mathbf{w}; \mathbf{X}, \mathbf{y}) + \alpha \, penalty(\mathbf{w})$$

  - α: balance parameter

- *loss*, for classification or regression:
  - Least squares loss
  - Hinge loss
  - Logistic loss
  - ...

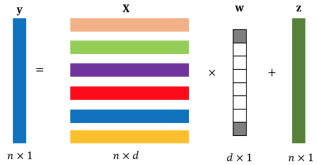
- *penalty*:
  - $\|\mathbf{w}\|_0$ seeks optimal features
  - However, it is not a valid norm, nonconvex and NP-hard
  - It is often relaxed to $\|\mathbf{w}\|_1$ (Lasso), which is the tightest convex hull

Lasso (Tibshirani, 1996)

- Based on ℓ_1-norm regularization on weight

$$\min_{\mathbf{w}} \; loss(\mathbf{w}; \mathbf{X}, \mathbf{y}) + \alpha \|\mathbf{w}\|_1$$

- In the case of least square loss with offset value, it looks like this ...

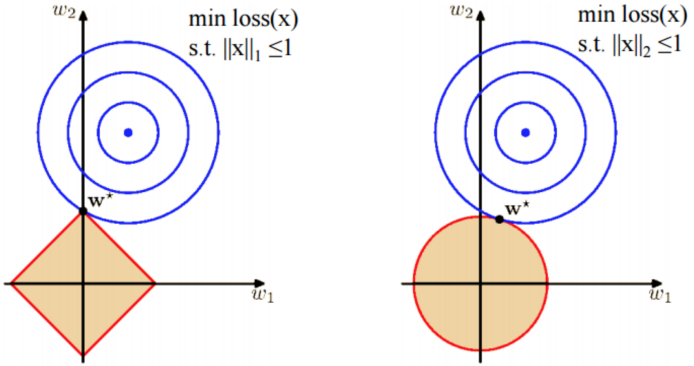


The feature score of the i-th feature is $|\mathbf{w}_i|$; the higher the value, the more important the feature is

- It is also equivalent to the following model

$$\min_{\mathbf{w}} \; loss(\mathbf{w}; \mathbf{X}, \mathbf{y}) \;\; \text{s.t.} \;\; \|\mathbf{w}\|_1 \leq t$$

Sparsity and norms



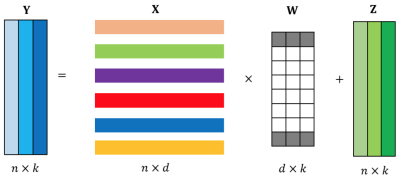
[Bishop, 2006],
[Hastie et al., 2009]

Extension to Multi-Class or Multivariate Problems

- Require feature selection results to be consistent across multiple targets in multi-class classification or multivariate regression

$$\min_{\mathbf{W}} \; loss(\mathbf{W}; \mathbf{X}, \mathbf{Y}) + \alpha \|\mathbf{W}\|_{2,1}$$

- $\|\mathbf{W}\|_{2,1}$ achieves joint feature sparsity across multiple targets
- In the case of least square loss with offset, it looks like this



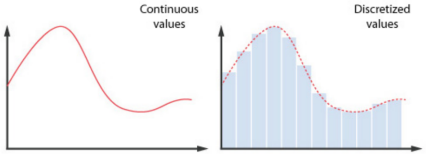
The feature score of the i-th feature is $||\mathbf{W}_{i*}||_2$; the higher the value, the more important the feature is

Sparse Learning based Methods - Summary

- Other sparse learning based methods
  - Multi-label informed feature selection (Jian *et al.*, 2016)
  - Embedded unsupervised feature selection (Wang *et al.*, 2015)
  - Adaptive structure learning feature selection (Du and Shen, 2015)
- Pros:
  - Obtain good performance for the underlying learning method.
  - Good model interpretability.
- Cons:
  - The selected features may not be suitable for other tasks.
  - Require solving computationally expensive non-smooth optimization problems.

Statistical methods

- This family of algorithms are based on different statistical methods to measure feature importance
- Most of them are filter feature selection methods
- Most algorithms evaluate features individually, so the feature redundancy is inevitably ignored
- Most algorithms can only handle discrete data, the numerical features have to be discretized first



T-Score (Davis and Sampson, 1986)

- It is used for binary classification problems
- Assess whether the feature makes the means of samples from two classes statistically significant
- The t-score of each feature f_i is

$$t_score(f_i) = \frac{|\mu_1 - \mu_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

  - μ_1, μ_2: mean value of samples from the first / second class
  - σ_1, σ_2: standard deviation value for samples from the first / second class
- The higher the t-score, the more important the feature is

Chi-Square Score (Liu and Setiono, 1995)

- Utilize independence test to assess if the feature is independent of class label
- Given a feature with r values, its feature score is

$$Chi_square_score(f_i) = \sum_{j=1}^{r}\sum_{s=1}^{c}\frac{(n_{js}-\mu_{js})^2}{\mu_{js}}, \qquad \mu_{js} = \frac{n_{*s}\, n_{j*}}{n}$$

  - n_{js}: # instances with the j-th feature value and in class s
  - n_{j*}: # instances with the j-th feature value
  - n_{*s}: # instances in class s
- Higher chi-square indicates that the feature is more important

Statistical based Methods - Summary

- Other statistical based methods
  - Low variance – CFS (Hall and Smith, 1999)
  - Kruskal Wallis (McKnight and Najab, 2010)...
- Pros:
  - Computational efficient
  - The selected features can be generalized to subsequent learning tasks
- Cons:
  - Cannot handle feature redundancy
  - Require data discretization techniques
  - Many statistical measures are not that effective in high-dim space

Other Types of Methods

- Reconstruction based Feature Selection
  - Minimize reconstruction error of data with selected features
  - Reconstruction function can be both linear and nonlinear
- Hybrid Feature Selection
  - Construct a set of different feature selection results
  - Aggregate different outputs into a consensus result

Feature Selection with Structured Features: A Framework

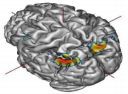
- A popular and successful approach is to minimize the fitting error penalized with structural regularization

$$\mathbf{w} = \arg\min_{\mathbf{w}} \; loss(\mathbf{w}; \mathbf{X}, \mathbf{y}) + \alpha \, penalty(\mathbf{w}, \mathcal{G})$$

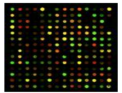
  - $\mathbf{w}$: feature weight; α: balancing parameter
  - The structure among features is encoded in $\mathcal{G}$
- The above formulation is flexible in incorporating various types of structures among features

Group Structure – Group Lasso

- Features form group structure in many applications



brain region



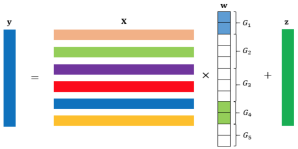
functional group

	group			
A	1	0	0	0
T	0	1	0	0
C	0	0	1	0
G	0	0	0	1

categorical variable

[Liu et al. SDM 2010]

- Group lasso selects or does not select a group as a whole

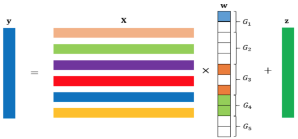


$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \alpha \sum_{i=1}^{k} h_i \|\mathbf{w}_{G_i}\|_2$$

- (Yuan and Lin, 2006): paper

Sparse Group Lasso

- For certain applications, it is desirable to select representative features from selected groups
- Sparse group lasso performs group selection and feature selection simultaneously

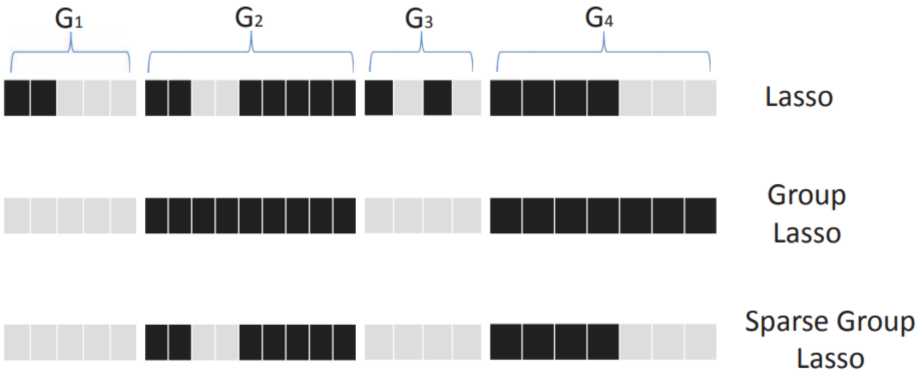


$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \alpha\|\mathbf{w}\|_1 + (1-\alpha)\sum_{i=1}^{k} h_i \|\mathbf{w}_{G_i}\|_2$$

- Joint group selection and feature selection
- (Friedman *et al.*, 2010): paper

Group Structure - Summary

- Comparison between Lasso, Group Lasso and Sparse Group Lasso



Feature Selection with Structured Features - Summary

- Incorporate feature structures as prior knowledge

Feature Selection with Structured Features - Summary

- Incorporate feature structures as prior knowledge
- Pros:
  - Improve the learning performance in many cases
  - Make the learning process more interpretable

Feature Selection with Structured Features - Summary

- Incorporate feature structures as prior knowledge
- Pros:
  - Improve the learning performance in many cases
  - Make the learning process more interpretable
- Cons:
  - Require label information to guide feature selection
  - Require to solve complex non-smooth optimization problems

Personalized Feature Selection

- Content information of nodes are highly idiosyncratic
  - E.g., blogs, posts and images of different users could be diverse and with different social foci
  - E.g., the same content could convey different meanings: "The house prices are getting higher and higher!": Seller – *positive*, Buyer – *negative* [Wang et al 2016]
- But nodes share some commonality to some extent
- How to tackle the idiosyncrasy and commonality of node features for learning such as node classification?

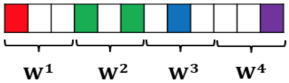
- (Li *et al.*, 2018): paper



Personalized Feature Selection

$$\min_{\tilde{\mathbf{W}},\mathbf{W}_i} \sum_{i=1}^{n} \|\mathbf{x}_i(\underbrace{\tilde{\mathbf{W}}}_{\text{global}} + \underbrace{\mathbf{W}_i}_{\text{local}}) - \mathbf{y}_i\|_2^2 + \beta \sum_{i=1}^{n} \underbrace{\|\mathbf{W}_i\|_{2,1}^2}_{\text{exclusive group lasso}} + \gamma\|\tilde{\mathbf{W}}\|_{2,1}$$

- $\tilde{\mathbf{W}}$: global feature weight for all nodes; $\mathbf{W}_i$: local feature weight for the i-th node
- To find personalized features, the goal is feature sparsity within each local feature weight
- The exclusive group lasso encourages intra-group competition but discourages inter-group competition



Personalized Feature Selection

- Local weights are clustered into groups to reduce overfitting

$$\min_{\mathbf{W}} \sum_{i,j=1}^{n} \mathbf{A}_{ij} \|\mathbf{W}_i - \mathbf{W}_j\|_F$$

  - Make connected nodes borrow strength from each other

- The objective function

$$\min_{\tilde{\mathbf{W}}, \mathbf{W}_i} J(\tilde{\mathbf{W}}, \mathbf{W}_i) = \sum_{i=1}^{n} \|\mathbf{x}_i(\tilde{\mathbf{W}} + \mathbf{W}_i) - \mathbf{y}_i\|_2^2 + \alpha \sum_{i,j=1}^{n} \mathbf{A}_{ij} \|\mathbf{W}_i - \mathbf{W}_j\|_F + \beta \sum_{i=1}^{n} \|\mathbf{W}_i\|_{2,1}^2 + \gamma \|\tilde{\mathbf{W}}\|_{2,1}$$

- (Li *et al.*, 2018): paper

Summary

- Feature selection is effective to tackle the curse of dimensionality and is essential to many data mining and machine learning problems
- The objectives of feature selection include
  - Building simpler and more comprehensive models
  - Improving learning performance
  - Preparing clean and understandable data
- Feature selection is equally important in the age of deep learning and big data
- A structured overview of feature selection from a data perspective
  - Feature selection for conventional data (four main categories)
  - Feature selection with structured features
  - Feature selection with heterogeneous data
  - Feature selection with multisource data

Quiz/Poll

What is a redundant feature?

- (A) a feature that has no relationship with the target
- (B) a feature that duplicates information in other features
- (C) a feature with many missing values that must be imputed first
- (D) the feature most strongly correlated with the target

Quiz/Poll

When can we use information theoretic methods for feature selection?

- (A) only when the features are continuous and no labels are available
- (B) only when the downstream model is linear
- (C) only after training a deep neural network
- (D) with labels and discrete (or discretized) features

Resources

- Materials from the previous years
- An introduction to Variable and feature Selection
- Blog on feature selection
- More on group Lasso
- Feature Selection for Data and Pattern Recognition
- Computational Methods of Feature Selection
- A Survey of Feature Selection Techniques

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